Effects of soil freezing on fine roots in a northern hardwood forest

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Abstract: We reduced early winter snowpack in four experimental plots at the Hubbard Brook Experimental Forest in New Hamphire for 2 years to examine the mechanisms of root injury associated with soil freezing. Three lines of evidence suggested that direct cellular damage, rather than physical damage associated with frost heaving, was the principal mechanism of root injury: (*i*) decreases in root vitality were not greater on sites with more frost heaving, (*ii*) in situ freezing damage was confined to first- and second-order roots in the organic horizons rather than entire root systems, and (*iii*) tensile strength of fine roots was not significantly compromised by experimental stretching to simulate ice lens formation. Although significant differences in the intensity of soil freezing (depth, rate, and minimum temperature) were observed across the plots, no clear effects of soil freezing intensity on root injury were observed. Snow manipulation had no effect on mycorrhizal colonization of sugar maple (*Acer saccharum* Marsh.) roots. A significant increase in root growth was observed in the second summer after treatments, coincident with a significant pulse of soil nitrate leaching. Through their effects on fine roots, soil freezing events could play an important role in forest ecosystem dynamics in a changing climate.

Résumé : Pendant 2 ans, nous avons réduit la couverture de neige au début de l'hiver dans quatre parcelles expérimentales à la forêt expérimentale de Hubbard Brook, au New Hampshire, pour étudier les mécanismes responsables des dommages aux racines causés par le gel dans le sol. Trois types d'indices ont indiqué que des dommages directs aux cellules, plutôt que des dommages physiques dus au soulèvement par la glace, étaient le principal mécanisme responsable des dommages aux racines : (*i*) la diminution de la vitalité des racines n'était pas plus grande dans les stations où le soulèvement par le gel était plus prononcé, (*ii*) les dommages in situ causés par le gel étaient limités aux racines de 1^er^ et 2^e^ ordres dans les horizons organiques plutôt qu'à l'ensemble du système racinaire et (*iii*) la résistance à la traction des racines fines n'était pas significativement compromise par un étirement expérimental pour simuler la formation de lentilles de glace. Bien que nous ayons observé des différences significatives dans l'intensité du gel dans le sol (profondeur, taux et température minimum) dans l'ensemble des parcelles, nous n'avons constaté aucun effet évident de l'intensité du gel dans le sol sur les dommages aux racines. La manipulation du couvert nival n'a eu aucun effet sur la colonisation des racines de l'érable à sucre (*Acer saccharum* Marsh.) par les mycorhizes. Une augmentation significative de la croissance des racines a été observée durant le deuxième été après les traitements. Cette augmentation coïncidait avec une poussée importante de lessivage du nitrate dans le sol. Par leurs effets sur les racines fines, les épisodes de gel dans le sol pourraient jouer un rôle important dans la dynamique des écosystèmes forestiers soumis aux changements climatiques.

[Traduit par la Rédaction]

Introduction

The importance of the winter season to biological processes in forest soils of northern regions has been demonstrated by studies of nutrient cycling activities beneath the snowpack (Campbell et al. 2005). Alteration of winter soil conditions affects root and microbial processes that feedback upon biogeochemical cycles and forest health (Groffman et al. 2001). For example, soil freezing events have been linked to spring pulses of nitrate in stream water (Likens and Bormann 1995; Mitchell et al. 1996; Fitzhugh et al. 2001, 2003) and to decline of sugar maple (*Acer saccharum* Marsh) in northeastern North America (Auclair et al. 1992; Pilon et al. 1994; Boutin and Robitaille 1995). The effects of soil freezing on fine roots are thought to be central to these responses (Groffman et al. 2001). Better understanding of the interaction between soil freezing and winter ecology of tree roots is necessary to predict changes in forest nutrient cycling and tree species composition likely to result from climate change.

Because snow insulates the soil, the timing of snowfall is an important determinant of the severity of soil freezing. Soil with a deep snowpack early in the season may remain unfrozen throughout a winter with severely cold air temperatures (Stadler et al. 1996; Shanley and Chalmers 1999; Decker et al. 2003). Conversely, the insulating properties of

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Table 1. Description of four study sites located in Hubbard Brook Experimental Forest, Woodstock, New Hamphire.

Site and treatment	Elevation (m)	Aspect	Slope (%)	Organic horizon depth (cm)	Stoniness (%)	Tree species in order of dominance
L1						
Control	380	105	9	5–8	15	Beal, Acpe, Fagr, Piru
Freeze		90	10	6–8	15	Beal, Acsa, Acpe, Fagr, Piru
L2						
Control	480	190	35	11–19	36	Acsa, Fram, Beal, Fagr
Freeze		180	50	4–9	20	Acsa, Beal, Fagr
H1						
Control	755	352	12	3–14	5	Beal, Acsa, Fagr, Acpe, Piru
Freeze		360	9	4–8	5	Acsa, Beal, Piru
H2						
Control	790	350	20	3–10	5	Beal, Acsa, Piru
Freeze		360	20	6–10	0	Beal, Acsa, Piru, Fagr

Note: Sites L1 and L2 are low-elevation sites, and H1 and H2 are high-elevation sites. Tree species are abbreviated using the first two letters of the genus and specific epithets: Acpe, striped maple; Acru, red maple; Acsa, sugar maple; Beal, yellow birch; Fagr, American beech; and Piru, red spruce.

snow allow the soil to remain frozen for the duration of the winter if the soil freezes prior to significant snowfall (Goodrich 1982; Hardy et al. 2001). In addition, snowpack greatly moderates fluctuations in soil temperature and may decrease the number of freeze–thaw events (Sutinen et al. 1996). Climate models currently predict an increase in winter precipitation for northeastern North America with more freeze–thaw events and delayed snow accumulation. This change could lead to more frequent and severe soil freezing events (IPCC 2001).

Ice formation in soil may affect fine roots in a number of ways including damage to cell membranes by ice crystal formation or desiccation, abrasion of root tissue, severing of root segments, or disruption of mycorrhizal associations (Sutinen et al. 1996; Bray et al. 2000). All of these interactions could lead to root death and impaired root uptake in the spring after soil thaw, thereby contributing to the elevated nitrate losses observed after deep soil freezing events (Mitchell et al. 1996; Groffman et al. 2001). In a previous snow-manipulation study, significantly higher root mortality (Tierney et al. 2001) and marked increases in hydrologic losses of nitrate (Fitzhugh et al. 2001) were observed in plots where snow had been removed to induce soil freezing; however, the mechanisms explaining these responses were not clearly established. Moreover, soil freezing was not very severe in that study, because it was conducted during relatively warm winters (Hardy et al. 2001). The present study was undertaken to provide additional evidence about the response of fine-root dynamics to soil-freezing events. We hoped to clarify the possible role of physical damage to roots by ice-lens formation and the response of physiological root vitality and subsequent root growth to differing rates and intensities of soil freezing. Our aim was to induce differing intensities of soil freezing by snow removal on plots at different physiographic locations (elevation and aspect) within the Hubbard Brook Experimental Forest (HBEF), New Hampshire. We hypothesized that the effects of soil freezing on root vitality would be greater in surface soil organic horizons and lower order (first and second) fine roots than for mineral horizons and higher order (third and fourth) fine roots. We also anticipated more severe impacts on root vitality with more rapid or severe soil freezing events. Finally, we expected a compensatory response of fine root growth in the summer following soil freezing events.

Materials and methods

Study site and experimental approach

The HBEF is situated in the White Mountain National Forest in central New Hampshire (43°56′N, 71°45′W). The forest is mature northern hardwoods with the major overstory species in our plots being sugar maple, yellow birch (*Betula alleghaniensis* Britt.), red spruce (*Picea rubens* Sarg.), American beech (*Fagus grandifolia* Ehrh.), white ash (*Fraxinus americana* L.), and red maple (*Acer rubrum* L.) (Table 1). Important understory species include striped maple (*Acer pensylvanicum* L.) and hobblebush (*Viburnum lantanoides* Michx.). The snowpack typically persists at the HBEF from mid-November to mid-April (165 days, 30 year mean) with January air temperatures averaging –9 °C and winter temperatures averaging –4.7 °C (Hardy et al. 2001). Site aspect, elevation, and tree composition are important factors influencing soil frost in the region with north-facing, mixedwood stands experiencing deeper frost than south-facing hardwood stands (Shanley and Chalmers 1999).

Our experimental design consisted of eight 12 m × 12 m plots (Table 1). The plots were located within four sites, two south-facing sites at low elevations (L1 and L2, 380 and 480 m, respectively) and two north-facing sites at high elevations (H1 and H2, 755 and 790 m, respectively), with one snow reduction treatment (regularly cleared of snow) and one reference (control) plot at each site. The objective was to induce different rates, intensities, and depths of soil frost in the plots. In both shoveled and reference plots, understory shrubs were clipped to facilitate shoveling in the treated plots. Snow was removed using snow shovels after each snow event in which a minimum of 5 cm of snow accumulated. To protect the forest floor and monitoring equipment and to preserve winter albedo on the plots, 5 cm of snow remained on the plots through the shoveling period. Plots were shoveled until late January in 2003 and 2004, after which snow was allowed to accumulate on all of the plots.

Winter environmental measurements

Soil temperature was measured in each plot using a Campbell Scientific CR 101x datalogger with a vertical string of thermistor probes inserted horizontally at 10 cm intervals starting at 0 cm. The datalogger system recorded temperature readings every minute and stored hourly means. Unfortunately, one site, (H2) encountered chronic electronic problems resulting in extensive intervals without accurate data. Frost depth was measured throughout the winters of 2002–2003 and 2003–2004 using frost tubes with two replicate tubes in each plot. The frost tubes were constructed from flexible PVC tube (1.27 cm inside diameter) filled with methylene blue dye (Ricard et al. 1976). The flexible tube was inserted into a polyethylene casing to a soil depth of 50 cm. To measure frost depth, the dye-filled tubes were removed from the rigid casing and the length of frozen dye (clear and solid as opposed to dark blue and liquid) was measured with a ruler. In both years, these measurements were taken at approximately weekly intervals from early December until soil thaw in April and May.

Frost heave of the soil surface was measured at times of maximum frost penetration at each of the plots via level survey. The elevations of eight equally spaced points in each plot were measured relative to unfrozen levels measured in October 2003. The points were marked by 10 cm long nails placed into the ground such that the head was at the surface. The frost heave reported is the mean of all eight points. A benchmark point was established by driving a 15 cm long nail into a large root of a mature tree, the standard practice of road survey crews; these roots strongly resist frost heaving. At the time of the elevation surveys, we collected two soil cores from the treatment plots (with a diamond-coated core barrel used for sampling concrete) and one soil core from the reference plots. These cores were transported to cold rooms, and the outer surface of each core was visually examined for ice lenses and physical damage to roots.

Sample collection and root damage experiment

Three replicate blocks of soil were removed from the plots in early spring of 2003 (21 April – 12 May) and 2004 (15 April – 10 May) when the soil temperature was 5 ± 2 °C (mean ± range). Soil blocks were divided into organic and mineral horizons in the field, placed in plastic bags, and refrigerated at 4 °C until roots were processed (within 24 h of collection). Fine roots (<2 mm in diameter) were separated from the soil, washed with cold running water, and placed on a glass dish over ice. First- and second-order roots were separated from third- and fourth-order roots using a razor blade. This step in the process was important, because first- and second-order roots are functionally most similar among species (Comas and Eissenstat 2004).

Assay of root vitality

Triphenyl-tetrazolium chloride (TTC) assays were used to compare fine root vitality between treated and reference plots (Comas et al. 2000). The TTC assay provides a quantitative measure of potential root respiration and reflects the number of living cells per unit root dry mass (Ruf and Brunner 2003). First, the root samples were chopped into 1–2 mm pieces with a razor blade, placed in separate containers, and refrigerated briefly until weighing. External water was completely removed from the roots by repeated patting of the root sample with tissue paper. The roots were then split into two subsamples, and 0.20–0.30 g of root (moist mass) was weighed into 30 mL glass scintillation vials to assure a dry mass of about 0.10 g. One subsample was used to obtain water content of the roots (calculated as the difference between the moist mass and the mass after drying in a 60 °C oven for 48 h), and the other was used for the root vitality analysis. Ten millilitres of a 0.1 mol/L potassium phosphate buffer (pH 7.0) solution containing 0.6% TTC and 0.05% Tween 20 was added to each root vitality sample vial (Ruf and Brunner 2003).

Samples were placed in a vacuum (3.0 MPa) for 45 min to draw solution into the root cells. Vials were capped and incubated at 30 °C for 15 h. The TTC – phosphate buffer solution was then drained off, and roots were rinsed twice with double distilled water and excess water was removed with a pipette. The samples were ground in 10 mL of 95% ethanol with sterile sand and a mortar and pestle. Samples were centrifuged for 1 min at 70 r/min and 5 mL of the supernatant was run on a spectrophotometer (GeneSys 40 portable; Thermo Electron Corp., Waltham, Mass.), and absorbance at 520 nm was recorded. A tube with 95% ethanol was used as a blank. Reduction of TTC was calculated as absorbance (520 nm) per gram of root dry mass. Absorbance per gram of root dry mass was the response variable used in a linear mixed model run in SPSS version 14.0 (SPSS Inc., Chicago, Ill.). For the soil block data, a full factorial model included main fixed effects of year (2003 or 2004), site elevation (high or low), treatment (reference or shoveled), soil horizon (organic or mineral), and root order (first and second or third and fourth) and random effects of site, site × treatment interaction, site × treatment × soil block replicate interaction, and site × treatment × soil block replicate × soil horizon interaction. Significant interactions were further explored using the estimated means comparison to obtain post hoc significance levels.

Root growth by minirhizotron

To track changes in growth of fine tree roots, we installed four minirhizotron tubes (5.08 cm inside diameter) made of cellulose acetate butyrate (CAB) in each of the plots. Tubes were installed October 2000 or May 2001 at a 45° angle to the depth of obstruction by rocks (56–71 cm of tube length, 40–50 cm vertical depth). Starting in October 2002, images were taken from the tubes monthly from May to October. An indexing handle (Bartz Technology Inc., Santa Barbara, Calif.) allowed us to return to the same exact locations at contiguous 12.5 mm intervals down the tubes. This approach resulted in 100–110 images (12.5 mm × 18 mm) per tube per filming date. Images were captured from the camera to a laptop computer in the field using the BTC I-CAP program (Bartz Technology Inc., Santa Barbara, Calif.). For each location, a chronosequence of 12 images (October 2002–2004) was scored for the total number of roots, the number of new root tips, and the number of previously recorded roots that had grown since the previous measurement.

The measure of root growth (RG) presented here consists of the number of new root segments divided by the total root segments in a frame over time. This proportional RG was analyzed by linear mixed model with filming date, treatment, and soil depth interval as fixed effects in a full factorial model and site, site × treatment, site × treatment × tube, and site × treatment × tube × depth as random effects. Across sites, dates, depths, and treatments, frames had 2.79 ± 0.14 (mean ± SE) new root segments and 16.88 ± 0.47 total root segments per time. Seasonal proportional RG for 2003 and 2004 (i.e., the total number of new root segments in a growing season (May–September) divided by the mean number of old and new root segments in that chronosequence for the year) was analyzed for the effects of year, site, soil depth interval, and treatment using this factorial model. Previous detailed studies at the HBEF indicated that root growth estimates were only moderately sensitive to whether root number, length, or mass was used in the calculations (Tierney and Fahey 2001); the large number of minirhizotron images (over 38 000) precluded tracing of root length in this study.

Ice lens mimic experiment

To quantify the effect of physical damage to roots from frost heaving, we designed an experiment to mimic the displacement imposed upon a root by ice-lens formation and consequent changes in the tensile strength of roots. Based on measurements of ice lenses in soil cores drilled from the plots in late January – early February 2003 (see above), a mean ice lens caused 2 mm displacement. Moreover, many of the ice lenses that we observed formed directly beneath fine roots. We collected winter-hardened sugar maple roots from trees near the two low-elevation plots in January 2004. Frozen soil blocks were thawed slowly in a 2 °C refrigerator, and 10 cm lengths of second-order roots were chosen. The 10 cm lengths were cut in half to create root pairs for the experimental treatment to reduce variation in tensile strength between root samples.

The tensile strength of a segment that was stretched while frozen (to mimic ice lens formation effects) was compared with the tensile strength of its pair segment that was only frozen. The ends of the root were protected and widened by epoxy glue. Once the epoxy had hardened, the root segments were frozen at –10 °C for 48 h. Segments used had diameters of 0.86 ± 0.27 mm (mean ± SD) and spans (length between epoxied ends) of 11 ± 3.1 mm.

To mimic displacement by an ice lens, one root segment of each pair was bent using a 2 mm diameter metal cylinder lowered at a constant rate (1 mm/min) by an Instron material testing system (model 4400; Instron, Norwood, Mass.) to a total displacement of 2 mm. The root was kept cold by placing dry ice in a tray beneath the root. All root segments were allowed to come to room temperature, and tensile strength tests were performed to compare stretched and unstretched roots. Over 40 root pairs were tested; however, after analysis of displacement and stress-strain charts, only five pairs of roots yielded data that were not influenced either by slippage or by premature breaking of the root at the clamp points. Differences in the tensile strength (Newtons of force at breaking point) were compared by two-tailed paired t test between stretched and unstretched roots.

Mycorrhizal colonization

In May 2004, three replicate samples of first- and second-order sugar maple roots were collected from each 10 m × 10 m plot. Roots were washed and stored in 50% ethanol solution until clearing and staining. Roots were cleared using successive immersion in 10% KOH solution at 70 °C. Five 2 cm root sections were included in each vial (two vials and 10 root sections per soil block), which received 10 mL each time the solution was changed. The solution was changed every 30 min until the KOH was clear. The total period ranged from 1–7 h. Cleared roots were rinsed with double deionized (DDI) water and acidified with 10 mL of 1% HCl for 1 h. Acidified roots were rinsed with DDI water and submersed in staining solution (50:50 glycerol:vinegar with Quink ink). Roots were kept in the stain overnight. The next morning, the roots were rinsed and placed in destaining solution (50:50, glycerol:vinegar). After at least 2 days in the destaining solution, the cortex was stripped from the roots using fine jeweler's forceps under a dissecting scope. This was necessary for two reasons: (*i*) to obtain a flat mount and (*ii*) light would not pass through the opaque suberized layer of mature tree roots. The cortices were flattened and placed uniformly on a slide, a few drops of destain solution were added, and a coverslip with 1 mm square grid was placed over the preparation. On each slide, at least 20 hits on the grid were scored for a total of 200 scores for each soil block. The percent colonization for each of four scored structures (arbuscules, vesicles, coils, and mycorrhizal hyphae) was analyzed by two-way ANOVA with site and snow removal as main effects.

Results

Soil temperature, frost, and heaving

As expected, soil frost penetrated earlier and more deeply in the soil at the high elevation (H1 and H2) than at the low elevation (L1 and L2) snow-removal plots in 2003 (Fig. 1). However, the maximum depth of frost penetration in H1 was similar to the two low-elevation plots by early February (about –35 cm); H2 had much deeper soil frost (Fig. 1). This trend was repeated in 2004: frost penetration was comparable in H1 and the two low-elevation plots (Fig. 1). Minimal soil frost penetration was observed in the reference plots in 2003. However, in the colder winter of 2004, frost penetrated to –5 to –15 cm depth in the reference plots by late January, which was still much less than in the treatment plots. Hence, although our aim to induce severe soil freezing was accomplished by the snow removal treatment, our expectation of consistently different soil freezing intensity in the high-elevation versus the low-elevation plots was only achieved for plot H2.

The patterns of frost penetration were generally reflected in the soil temperature measurements for three sites (L1, L2 and H1; Fig. 2). Soil temperatures were colder (Wilcoxin signed ranks; $p < 0.001$ for all but L1) and exhibited more variation in the shoveled than the reference plots at both the soil surface and –10 cm depth. Soil temperatures at the soil surface were more variable and colder than at –10 cm depth, and temperatures were markedly colder in 2003–2004 than 2002–2003. However, no consistent differences in soil temperature were observed between the low and high elevation plots (Fig. 2). Sporadic measurements from treatment plot H2 indicated that soil temperatures were consistently about

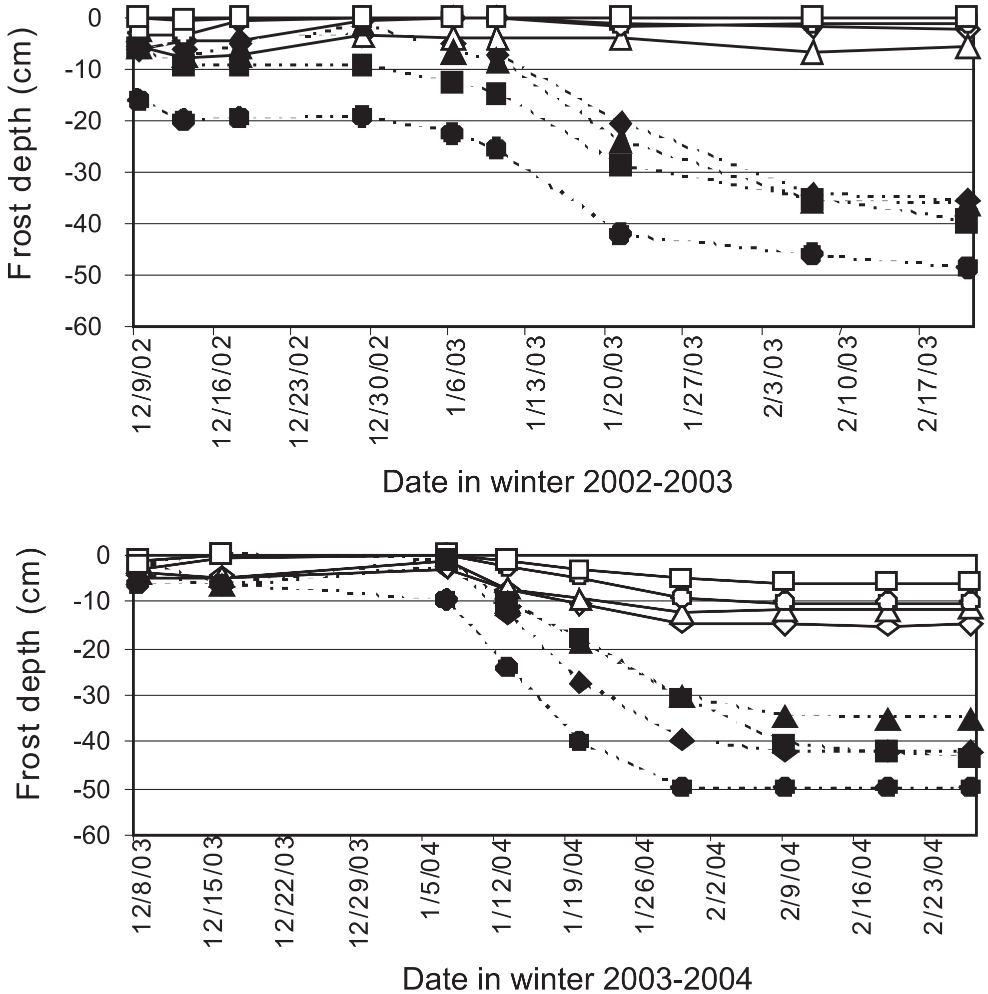


Fig. 1. Soil frost depths (mean value of two replicate frost tubes per plot) for two treatment winters at treatment (solid symbols, broken lines) and reference (open symbols, solid lines) plots for four sites: L1 (triangles), L2 (diamonds), H1 (squares) and H2 (circles).

1 °C colder than the other treatment plots from mid-January to mid-February 2004.

As an index of the rate of soil freezing and frost penetration, we quantified the time interval between the first full day of subfreezing soil temperature at 0 cm depth and that at 10 cm depth. This rate of temperature decrease differed both between years and among sites. In the first treatment year, the temperature of the soils in the L1 and H1 shoveled plots decreased much more rapidly (37 days to penetrate to 10 cm) than at H2 (70 days) and L2 (78 days), although H2 was frozen earliest (Fig. 1). In the second treatment year, the rates of temperature decrease in the top 10 cm were similar across all the treated plots (65 ± 2 days).

Using the frost tube data, slopes for the linear increase in frost depth in January of both years were compared between treatments and years. The depth of frost penetration differed for this period between sites (Fig. 1). As with the temperature data, shoveled plots froze significantly faster than control plots in both years ($F_{1,12} = 79.2$, $p < 0.001$). Between years, frost depth increased significantly faster on all treated plots in January 2004 ($F_{1,12} = 12.7$, $p = 0.004$). In both years, H2 had the steepest slope for frost penetration on treated plots (Fig. 1).

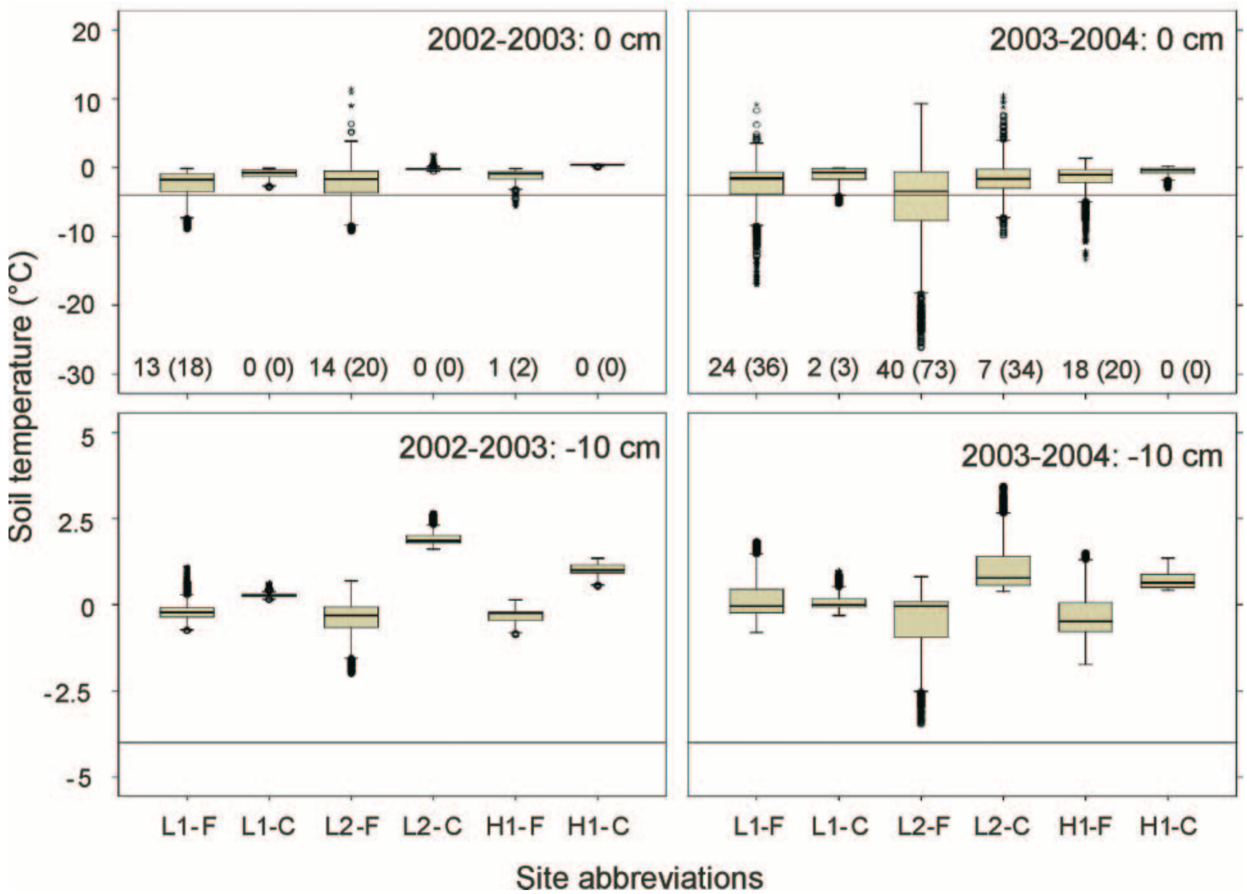
Differences in frost heave experienced on the plots showed that frost heaving was greatest on the treatment plots at L1 (41 mm ± 22; mean ± SD) and H1 (47 ± 13 mm). Frost heaving was the least on L2 (12 mm ± 16) and H2 (19 mm ± 15). Only L1 and H1 had significantly greater heave on treatment than on reference plots (paired t test; $p = 0.002$ and <0.001, respectively).

Root vitality response to snow removal treatment

The vitality of first- and second-order roots in the organic horizon was strongly reduced by the snow removal treatment in both years. In contrast, there was no significant response to the treatment in mineral soil roots or in third- and fourth-order roots (Fig. 3). Several higher order interactions involving differences between years were observed, generally reflecting higher vitality in 2003 than 2004, a pattern that was significant in some comparisons (Table 2). Site elevation did not affect root vitality or treatment response, which was not surprising because the elevation effect on soil freezing was not consistent (Fig. 1 and 2).

After the second year of snow removal treatment, there was no significant difference in mycorrhizal colonization for hyphae (39.3% ± 4.97%; mean ± SE), arbuscules (5.15% ± 1.00%), vesicles (7.24% ± 1.24%), or coils (7.40% ± 1.47%) in sugar maple roots from reference and

Fig. 2. Box plots of soil temperature during the soil freezing period for three sites: L1, L2, and H1 at treatment (F) and reference (C) plots. In winter 2002–2003, complete data for all sites during the freezing period were available for 70 days (1476 temperature recordings) and in 2003–2004 were available for 100 days (2310 temperature recordings) for each box plot. Numbers above the *x* axes in the 0 cm plots are the number of days below –4 °C for the mean daily temperature and for the minimum daily temperature (in parentheses). A reference line is shown for –4 °C because of its biological significance. In the box plots, the horizontal line is the median value, and the box shows the spread of 50% of the data. Error bars show the interquartile range (80% of the data). Circles are outliers (data values beyond three interquartiles), and asterisks are extreme values.



treatment plots (ANOVA: $F_{1, 18}$ = 0.203, 0.692, 1.207, and 0.039, respectively; all $p > 0.386$).

In laboratory experiments, the effects of stretching (to simulate ice lens damage) on the tensile strength of frozen roots was not significant (paired *t* test: p = 0.43); tensile strength of sugar maple roots that had been frozen (maximum load sustained 21.85 ± 7.55 N) was similar to that of roots that had been frozen and mechanically stretched (20.11 ± 5.81 N).

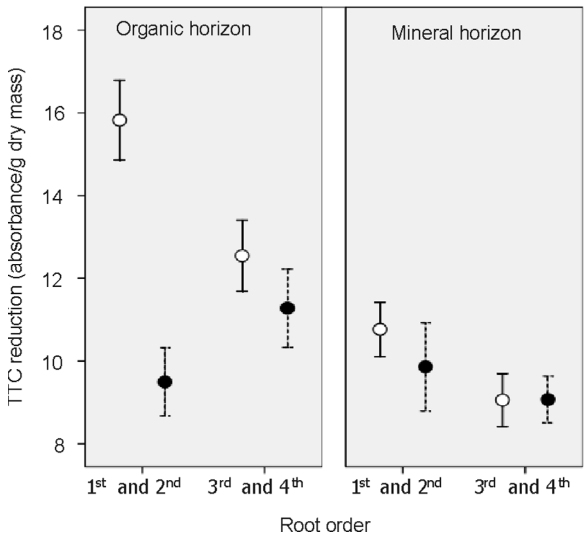
Root growth response to snow removal treatment

Proportional RG in the field was significantly higher in the treated than the reference plots on the first three filming days of 2004 (significant treatment × day interaction; $F_{10, 859}$ = 3.787, $p < 0.001$) (Fig. 4). When proportional RG was pooled for a season (May–September), there was a significant three-way interaction between year, site, and treatment effects ($F_{3, 6}$ = 6.779, p = 0.023). There was no significant difference in seasonal proportional RG between treatments at any of the sites in 2003 (treatment $F_{1, 70}$ = 0.180, p = 0.672) (Fig. 5). In 2004, overall annual root growth was higher in the treatment plots; however, most of this response occurred at two sites only: L2 and H1 (significant site × treatment interaction $F_{3, 71}$ = 4.477, p = 0.006) with marginally higher production at L1 and no difference in production at H2 (Fig. 5). As expected, RG was highest during early summer months and greater in the 0–10 cm depth than deeper in the soil. However, there was no significant treatment by depth interaction for proportional RG despite the strong root vitality response (Fig. 3).

Discussion

The interaction between soil freezing processes and mature forest tree roots has rarely been studied. Most studies of soil freezing effects on tree roots have been carried out with potted seedlings or in orchard settings with fruit trees. In this study, freezing damage at the cellular level was shown to be more significant than physical damage to root systems in three ways: (*i*) decreased vitality was limited to first- and second-order roots and not to entire root systems, as would be expected if physical damage to lateral roots prevailed; (*ii*) stretching frozen roots did not compromise the tensile strength of the root more than freezing alone; and (*iii*) decreases in root vitality were not greatest on the two

Fig. 3. Comparison of root cell vitality in spring by TTC vitality test as shown by three-way interaction (treatment × horizon × order): treatment (solid symbols, broken lines) and reference (open symbols, solid lines) plots in organic and mineral horizons for root order (first and second order vs. third and fourth order) of fine tree roots (<2 mm diameter). Values are means, and error bars are SEs (N = 12; 3 replicate samples from 4 plots). Higher absorbance (at 520 nm) indicates greater reduction of TTC to formazan and higher respiration potential of the cells.



sites with greatest frost heaving (L1 and H1). Results from this study strongly suggest that soil freezing damage occurred as cellular injury to first- and second-order roots in the organic horizon (Fig. 3). Decreased vitality when measured by TTC is indicative of decreased potential respiration (Comas et al. 2000) and, on a dry mass basis, reflects the number of living cells (Ruf and Brunner 2003). Visual assessments of roots in 2003 and 2004 during root processing support this interpretation, because cells were much less turgid in roots from shoveled plots, perhaps reflecting dessication damage.

Although our aim to induce consistently different rates and intensities of soil freezing between high-elevation and north-facing and low-elevation and south-facing plots was unsuccessful, site differences did provide some useful insights. The clearest site differences in root response to the treatment were decreased vitality of first- and second-order roots and increased fine root production in the L2 and H1 plots. The increased summer root growth following soil freezing matches previous results (Tierney et al. 2001) and, presumably, indicates a compensatory response to reduced root uptake activity. These two plots (L2 and H1) also exhibited a significantly greater response of nitrate leaching to the treatments than the other plots (S. Fashu-Kanu and C.T. Driscoll, Department of Civil and Environmental Engineering, Syracuse University, Syracuse, N.Y., unpublished data). The cooccurrence of greater root responses and nitrate leaching lends support to the hypothesis that root damage by soil frost reduces nitrogen uptake and stimulates nitrate leaching. However, these two plots did not experience distinctive levels of soil frost penetration, frost heaving, or soil temperature. Indeed, their main commonality is the dominance of sugar maple (Table 1), a pattern which suggests that tree composition was important for root sensitivity to freezing in this study. In contrast, Tierney et al. (2001) did not observe significant differences in the response of root dynamics to soil freezing between sugar maple and yellow birch dominated plots at the HBEF. Notably, the depth and intensity of soil freezing was much less in that previous snow removal study, because winter temperatures were relatively mild (Hardy et al. 2001). Nevertheless, both root dynamics (Tierney et al. 2001) and nitrate leaching responded to those mild soil freezing events (Fitzhugh et al. 2001). In sum, these combined results support a connection between soil freezing, root damage, and nitrate leaching in northern hardwoods;however, some still unidentified factors, and not apparently the severity of soil frost (as measured herein), induces differences in the ecosystem responses (see also Fitzhugh et al. 2003).

No difference was observed in colonization of sugar maple roots by arbuscular mycorrhizae (AM) between control and treated plots. Work by McGonigle and Miller (1999, 2000) in agricultural systems has shown that extraradical mycelium (hyphal mass outside the root) survives in the soil over winter and rapidly recolonizes roots of annuals in the spring, and disturbance does not lower colonization in soils with high inoculum densities. Klironomos et al. (2001) have shown that AM fungal species of *Glomus* are most abundant in the winter, and their colonization rates are less affected by freezing. It would be quite informative to look for shifts in mycorrhizal fungal populations in future snow manipulation studies. The lack of freezing effect on AM colonization contrasts with recent work at the HBEF showing large differences in mycorrhizal colonization of sugar maple roots between reference and calcium-amended watersheds (Juice et al. 2006). This contrast adds support for the assessment that physical disturbance is less important than soil chemistry and plant nutrient status for levels of AM colonization (McGonigle and Miller 2000). An interesting question that arises is whether AM influence the ability of root cells to harden. Given the different structure of arbuscular versus ectomycorrhizal relationships, the heightened intimacy with the cell membrane in AM relationships may well have consequences for freezing tolerance of species with AM associations.

In this study, we were careful to differentiate root effects by soil horizon. There was no evidence of damage to roots in the mineral horizon, even though this horizon experienced considerable frost penetration (Fig. 1). Despite the penetration of frost into this horizon, thermal regimes were much more clement below 10 cm depth with little variability around median values and with no temperatures below –4 °C (Fig. 2). This might explain the lack of treatment response for roots in the mineral horizon; however, the exact temperature at which damage occurred to fine roots in this study is unknown. Future field studies should concentrate on freezing processes in the organic horizon and attempt to quantify rates of soil freezing and root dehydration.

When compared with fine roots in the mineral horizon, fine roots in the organic horizon have greater specific root length, higher percent branching, and shorter lifespans (Fahey 1992; Fahey and Hughes 1994; Eissenstat et al. 2000). Low-order roots in the organic horizon are critical to water and nutrient uptake. Their loss reduces the ability of

Table 2. Significant interactions in full factorial linear mixed model for root vitality comparing effects of year (2003 or 2004), site elevation (high or low), treatment (reference or shoveled), soil horizon (organic or mineral), and root order (first and second or third and fourth) as main effects in field plots at the Hubbard Brook Experimental Forest, New Hampshire.

			Post hoc analysis	
Source of variation	*F*	*p*	Comparison	*p*
Treatment (T) × order (O)	11.094	0.001	Shoveled plots: first and second order < third and fourth	<0.001
T × horizon (H)	5.651	0.030	Shoveled plots: organic horizon < mineral	0.001
Year × elevation × O	6.055	0.016	First- and second-order roots at high elevation: 2003 > 2004	0.021
T × H × O	4.404	0.039	First- and second-order roots in organic horizon: shoveled < reference	<0.001

Fig. 4. Proportional root segment production (as number of segments) in treatment (solid symbols, broken lines) and control (open symbols, solid lines) sites for two growing seasons following two treatment winters at three soil depth intervals. Values are means, and error bars are SEs (N = 16; 4 replicate tubes in 4 plots). Letters above the month labels denote the significance of post hoc comparisons of total root production (all depths) among filming dates. Months with the same letters were not significantly different at α = 0.05.

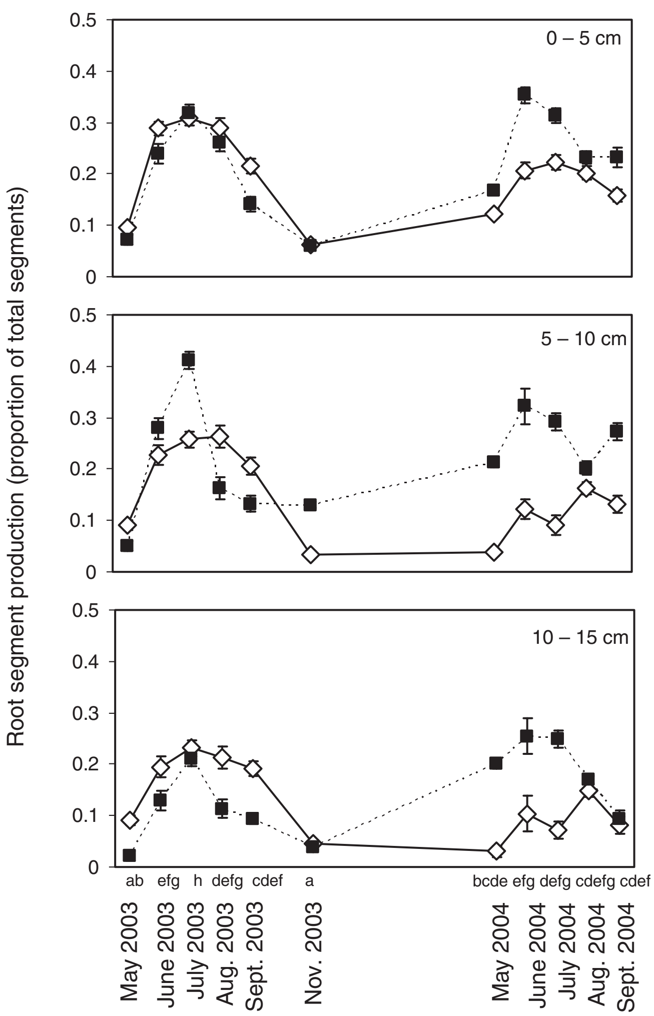
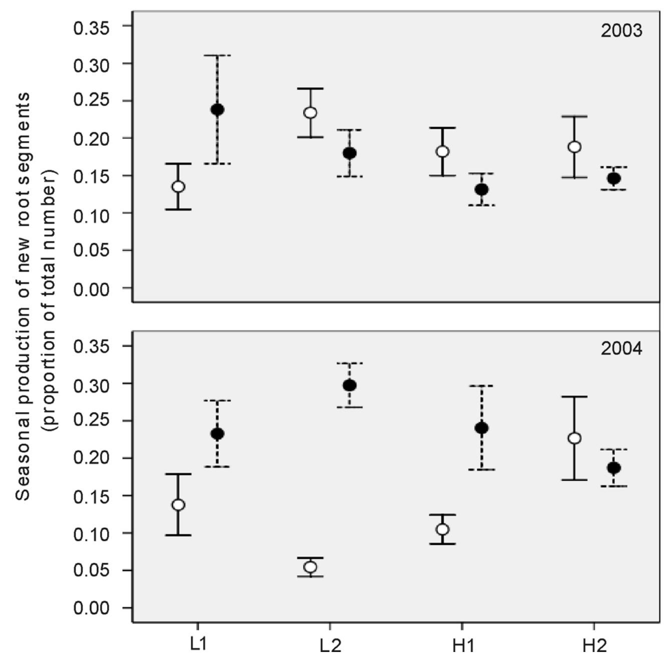


Fig. 5. Comparison of total seasonal new root segment appearance in the 2003 and 2004 seasons at four sites in snow removal (solid symbols, broken lines) and reference (open symbols, solid lines) paired plots. Values are means, and error bars are SEs (N = 4 replicate samples per plot).



trees to take up nutrients in early spring, and their replacement is necessary to support aboveground growth in the following season. Repeated freezing damage and new production of fine roots could represent a significant carbon drain on the tree. Root damage by freezing resulted in decreases in aboveground growth and increases in tree mortality in the following season for containerized conifer and birch seedlings (Dumais et al. 2002; Weih and Karlsson 2002; Zhu et al. 2002).

It remains unclear why the root growth response was only evident in the second year of treatment given that the decrease in vitality occurred in both years. More study is needed on the endogenous controls on root growth after root damage to mature trees (Tierney et al. 2003). The investment of carbon in new root growth must depend on a number of complex variables including tree carbon status, nutrient and water requirements, soil moisture, and soil nutrient concentrations. Carbon allocation to roots and, therefore, root growth is to some extent related to the previous season conditions, and root production has been strongly linked to temperatures in the month of October when trees begin the hardening process (Côté et al. 2003).

We should point out some methodological observations on root assays from this study. First, we attempted to use electrolyte leakage as an index of root damage (Levitt 1980). Unfortunately, this approach was not effective for our field study, because electrolytes had already been lost from damaged cells at the time of sample collection (same date as for TTC analysis). In contrast, the TTC method proved very reliable and, probably, will be the best choice available for future field studies of root vitality. We were careful to collect root samples in reference and treatment plots at roughly the same soil temperature rather than on the same day (i.e., soil warmed earlier on the reference than the treatment plots and in the low- than the high-elevation plots); this was necessary because the TTC assay is very sensitive to root growth and metabolic activity, which vary markedly with soil temperature. Also, we observed that results of the TTC assay differed between root species so that comparisons across species must be made with care.

Despite the inherent limitations of field experiments in terms of uncontrolled conditions and lower replication than laboratory or pot studies, such experiments are necessary for highlighting the complexity of interactions between soil freezing processes and tree roots. Winter root damage is suggested to be a critical factor explaining declines in several important tree species in the US Northeast including sugar maple (Auclair et al. 1992; Pilon et al. 1994; Boutin and Robitaille 1995). Soil temperature limitations on root growth are known to explain one of the most pronounced delineations of species distribution, the alpine treeline (Körner 1999). This work adds to the growing evidence that soil freezing events and their effects on roots may well play an important role in controlling tree distributions in northern hardwood forests and suggests several important areas for future research.

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